

Ravichander et al. (2019) proposed a benchmark for evaluating quantitative understanding in textual entailment. However, these datasets blend numerical reasoning with broader language understanding tasks, making it challenging to isolate numerical processing abilities.

Probing numerical understanding in LMs Several works have probed numerical comprehension in encoder models. Wallace et al. (2019) trained probing models to assess numerical understanding embedded in model representations, while Johnson et al. (2020) extend this conclusion to multi-language settings. Naik et al. (2019) used contrastive tests to evaluate models’ understanding of number magnitudes. Geva et al. (2020) demonstrated that finetuning on numerical reasoning data enhances the understanding. Unlike these studies, which focus on embeddings, our work emphasizes generating correct answers in autoregressive models. Recent efforts on such models include Razeghi et al. (2022), who studied few-shot learning correlations between term frequency and performance, and Zhang et al. (2024a), who identified key components in LLMs for basic arithmetic tasks. These works focus on some most classic tasks and our benchmark expands on these by incorporating diverse numerical representations, tasks, and digit ranges, offering a more comprehensive analysis.

Numerical dataset in specific domains Datasets like those proposed by Spithourakis & Riedel (2018) and Lin et al. (2020) test numerical commonsense reasoning, while others focus on specific contexts, such as financial reasoning (Chen et al., 2021; 2022) or tabular data (Akhtar et al., 2023). These works highlight numerical reasoning within specific domains rather than general numerical processing tasks. In contrast, our benchmark targets core numerical understanding, emphasizing tasks decoupled from domain-specific constraints.

Mathematical Reasoning Datasets Despite its close relationship with NUPA, *mathematical reasoning* is a broader field involving diverse skills such as task comprehension, equation solving, tool usage, and more (Lu et al., 2023b). While correct numerical processing is a critical component of mathematical reasoning, it is not the entirety of it (Stolfo et al., 2023). Datasets like MathQA (Amini et al., 2019), GSM8k (Cobbe et al., 2021), MATH (Hendrycks et al., 2021b), and SVAMP (Patel et al., 2021) focus on math word problems requiring multi-step reasoning and problem-solving. Few works isolate numerical processing from mathematical reasoning. Saxton et al. (2019) introduced a dataset for numerical tasks, such as adding floating-point numbers, but lacked task categorization by difficulty or length. Moreover, mixing numerical and algebraic tasks complicated analyses of pure numerical processing. Our benchmark addresses this gap, offering fine-grained categorization and evaluation of numerical understanding tasks.

6 CONCLUSION

We investigate NUPA of LLMs and introduce a comprehensive benchmark, the NUPA Test, to reveal that numerical problems remain challenging for modern LLMs. Our comprehensive test, which includes a variety of numerical representations and tasks, has exposed the surprising vulnerability of LLMs in this fundamental area. To explore ways to improve NUPA, we extend and evaluate previous pretraining techniques on the NUPA benchmark. While direct finetuning on the NUPA tasks does improve the performance, utilizing those tricks specifically designed for NUPA in the finetuning tends to harm NUPA, suggesting that these methods are not easily transferable to practical LLMs. We also explore the potential of chain-of-thought techniques to enhance NUPA and discuss their limitations.

7 LIMITATION

As a benchmark that specifically focuses on number understanding and processing abilities, we acknowledge that the range of tasks could still be incomplete and biased toward certain aspects. We will continue updating our benchmark, including but not limited to adding new tasks and refining existing ones to ensure appropriate difficulty. Additionally, the number of models we have tested so far is limited, and we plan to include more promising pretrained models in future evaluations.

On the other hand, although we have identified the limitations of LLMs’ NUPA, the existing solutions each have their own drawbacks. We have yet to find a path that fully addresses the problem. Solving this issue may require research across multiple fields, such as enhancing the diversity of pretraining corpora, developing new techniques, or enabling more efficient reasoning paradigms that make more complex CoT approaches feasible. We hope our work can contribute to and be complemented by advancements in these areas.

REPRODUCIBILITY STATEMENT

We have made every effort to ensure that the results presented in this paper are fully reproducible. Detailed descriptions of the number formats, construction and metrics of our NUPA dataset are provided in Section 2 and A.1.5, and examples for each task in A.4.4. To further facilitate reproducibility, we have included the complete dataset and the source code, enabling the generation of the entire dataset and the training and assessment of models, within the supplementary materials and the github page https://github.com/GraphPKU/number_cookbook. Researchers wishing to generate NUPA benchmark or replicate our experiments can refer to these resources for all necessary information.

ETHICS STATEMENT

In conducting this research, we have adhered to the highest ethical standards to ensure the integrity and fairness of our work. For source code releases, we have ensured compliance with applicable legal standards. During the construction of the dataset, all data was entirely generated randomly, without including any personal identity information or other private data of individuals.

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